

$$\text{Sigmoid}(x) = \frac{1}{1 + e^{-x}}$$

$$\sigma(x) = \frac{1}{1 + e^{-x}} \\ = (1 + e^{-x})^{-1}$$

apply the chain rule,

$$\begin{aligned} \frac{d}{dx} \sigma(x) &= -1(1 + e^{-x})^{-2} \cdot \frac{d}{dx} (1 + e^{-x}) \\ &= -(1 + e^{-x})^{-2} \cdot (-e^{-x}) \\ &= (1 + e^{-x})^{-2} e^{-x} = \frac{e^{-x}}{(1 + e^{-x})^2} \end{aligned}$$

$$= \sigma(x)(1 - \sigma(x))$$

$$\therefore \frac{d}{dx} \sigma(x) = \sigma(x)(1 - \sigma(x))$$

During backpropagation, output $y = \sigma(x)$, the gradient is

$$\frac{d}{dx} \sigma(x) = y(1 - y)$$